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Research Interests

- Investigate stellar feedback and its impact on star formation and quenching processes at sub-galactic scales using multi-band imaging and integral field spectroscopy.
- Study the interactions within the multiphase interstellar medium.
- Explore the microstructure of shock- or photon-ionized nebulae in the Milky Way and nearby galaxies to refine ionization models and improve hydrodynamic simulations.
- Develop and tailor advanced machine learning techniques for efficient analysis and interpretation of astronomical datasets.

Education

- **Doctor of Philosophy**, Astronomy
Department of Astronomy, Tsinghua University 2020 - Now
- **Bachelor of Science**, Astronomy
Department of Astronomy, Xiamen University 2016 - 2020

Publications

- **Jing, T.** and Li, C. (2025), Regression for Astronomical Data with Realistic Distributions, Errors, and Non-linearity. *AJ*, 170, 45, 10.3847/1538-3881/add891
- **Jing, T.** and Li, C. (2024), On the Origin of Quenched but Gas-rich Regions at Kiloparsec Scales in Nearby Galaxies. *ApJ*, 975, 17, 10.3847/1538-4357/ad7367
- **Jing, T.**, Li, C., Yan, R., Cheng, C., Zhang, W., Ji, X., Li, N., Wang, J., Wu, C., and Yuan, H. (2024), Criss-cross Nebula: A Case Study of Shock Regions with Resolved Microstructures at Scales of 1000 au. *ApJ*, 962, 51, 10.3847/1538-4357/ad11d1
- Ma, C., Sun, Z., **Jing, T.**, Cai, Z., Ting, Y.-S., Huang, S., and Li, M. (2025), Can AI Dream of Unseen Galaxies? Conditional Diffusion Model for Galaxy Morphology Augmentation. *arXiv e-prints*, arXiv:2506.16233, 10.48550/arXiv.2506.16233
- Guo, R., Li, C., Zhou, S., Li, N., **Jing, T.**, and Cheng, Z. (2025), Mapping Dust Attenuation at Kiloparsec Scales. II. Attenuation Curves from Near-ultraviolet to Near-infrared. *Research in Astronomy and Astrophysics*, 25, 065017, 10.1088/1674-4527/add673
- Sui, C., Zhao, X., **Jing, T.**, and Mao, Y. (2023), Evaluating Summary Statistics with Mutual Information for Cosmological Inference. *Machine Learning for Astrophysics*, 30, 10.48550/arXiv.2307.04994

Collaborations

- [Affordable Multiple Aperture Spectroscopy Explorer \(AMASE\) project](#) [🔗](#): a in-processing project that will pair 100 identical multi-fiber spectrographs to map the ionized gas in the Milky Way and nearby galaxies at the spatial resolution of half an arcminute and a spectral resolution of R=15,000.

I am in Science Team of AMASE and work for the HII region sample selection and construct mock data cubes from simulations. I am also leading a subproject to study microstructures in shocked regions.

Approved Proposals

- PMO Delingha Millimeter Telescope 2022 (30.5 hours awarded) PI
Title: Detecting ^{12}CO in the Criss-cross Nebula to study shock-molecular cloud collision.
- SITELE@CFHT 2022B (5.5 hours awarded) PI
Title: High resolution IFS on Criss-cross Nebula as a case study of microstructure of shock regions
- MeerKAT 2022 (7.5 hours awarded) PI
Title: High resolution HI observation on Criss-cross Nebula: the case study for shock-cloud interaction

- SITEELLE@CFHT 2024B (PI: Xiao Li, 5.6 hours awarded)

CoI

Title: Large FoV IFS on UGC1382 as a case study of quenched but HI-rich galaxies

Technical Skills

- **Programming Language:** Python, C, FORTRAN, JavaScript, Mathematica, $\text{\LaTeX}$
- **Ionization Model:** [MAPPINGS V](#)
- **Spectrum Analysis:** [pPXF](#), self-developed code
- **Simulation Post-processing:** [COLT](#), [CHIMES](#)
- **Machine Learning Packages:** [TensorFlow](#), [JAX](#), [PyTorch](#), [scikit-learn](#), [AutoGluon](#)

Selected Talks and Posters

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| ◦ The 12th Sino-German Meeting on Galaxy Formation
Poster + Flash talk, Chengdu, China | 2025 |
| ◦ Decoding Galactic Evolution through the Interplay of the Multi-Phase Inter-stellar Medium
Poster + Flash talk, Nagoya, Japan | 2025 |
| ◦ The 12th Symposium on Interstellar Physics and Chemistry
Oral talk, Zhangye, China | 2025 |
| ◦ The 27th Guoshoujing Symposium on Galaxies and Cosmology
Oral talk, Changsha, China | 2025 |
| ◦ AMASE Science Meeting
Oral talk, Xiangtan, China | 2025 |
| ◦ JWST Discussion Workshop
Invited talk, Given remotely | 2025 |
| ◦ The Ecosystem of Gas and Dust in Galaxies Near and Far
Oral talk, Chiang Mai, Thailand | 2024 |
| ◦ FAST Deep Sky Survey Annual Meeting 2024
Oral talk, Beijing, China | 2024 |
| ◦ THU Galaxy & Cosmology Seminar
Invited talk, Beijing, China | 2024 |
| ◦ The 11th Symposium on Interstellar Physics and Chemistry
Oral talk, Nanjing, China | 2024 |
| ◦ AMASE Science Meeting
Two oral talks, Beijing, China | 2024 |
| ◦ The 26th Guoshoujing Symposium on Galaxies and Cosmology
Oral talk and Poster, Suzhou, China | 2024 |
| ◦ Resolving Galaxy Ecosystems Across All Scales 2023
Oral talk, Hongkong, China | 2023 |
| ◦ ACAMAR Gas in Galaxies Workshop 2023
Oral talk and Poster, Perth, Australia | 2023 |
| ◦ LAMOST MRS-N and Star Formation Evolution Seminar
Oral talk, Xuyi, China | 2023 |
| ◦ SDSS Collaboration Meeting 2021
Oral talk, Given remotely | 2021 |
| ◦ The 23th Guoshoujing Symposium on Galaxies and Cosmology
Oral talk, Hangzhou, China | 2021 |